

RIXIN LIU

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RESEARCH INTERESTS

My research centers on one question: *How should we manage GPUs when they must serve more than one workload?* I approach it through elastic GPU memory management, fault isolation and recovery, and multi-model serving. I am broadly interested in building AI infrastructure for training, inference, and RL.

EDUCATION

Rice University

Ph.D. student in Computer Science
Advisor: Prof. Jiarong Xing

Houston, TX
Sep. 2025 – Present

South China University of Technology

B.E. in Computer Science and Technology

Guangzhou, China
Sep. 2021 – Jun. 2025

OPEN-SOURCE PROJECTS

kvcached | Project Lead & Maintainer

Virtualized, elastic KV cache for dynamic GPU sharing

May 2026 – Present
1.5k GitHub stars

- Lead ongoing development and maintenance of kvcached, which decouples virtual addresses from physical GPU memory for elastic KV cache allocation across colocated LLMs; integrates with **vLLM** and **SGLang**.
- Project benchmarks report **2–28× lower time to first token** when serving three Llama-3.1-8B models on one A100-80GB under workloads with intermittent peaks, versus static memory reservation.
- Deployed at scale by **Tencent**; also adopted by **Red Hat's Sardeenz**, which builds on kvcached for dynamic multi-model serving.

RouterArena | Open evaluation platform for LLM routers

143 GitHub stars

- Co-developed an automated evaluation framework and public leaderboard that benchmarks LLM routers across **9 domains and 44 categories** at three difficulty levels, measuring accuracy, cost, optimality, robustness, and latency.
- Grew the leaderboard into a community reference spanning **33 academic and commercial routers**; adopted by the **vLLM Semantic Router** project for router evaluation.

SELECTED RESEARCH

GPU Fault Resilience in NVIDIA MPS

First author

- Characterized GPU faults and their end-to-end handling to understand how one failing MPS process can terminate other processes sharing the GPU.
- Developed complementary mechanisms: memory-fault isolation in the open GPU kernel driver and fast recovery through virtual-memory-based sharing of GPU-resident state, restoring pre-fault throughput within **300 ms** in end-to-end LLM inference.

PUBLICATIONS & PREPRINTS

*Equal contribution.

Characterization-Guided GPU Fault Resilience in NVIDIA MPS

Rixin Liu, Xingqi Cui, Kaijian Wang, Xinheng Ding, Zirui Liu, Yuke Wang, Jiarong Xing.
Under review, 2026.

Deterministic Inference Across Tensor Parallel Sizes That Eliminates Training-Inference Mismatch

Ziyang Zhang*, Xinheng Ding*, Jiayi Yuan, **Rixin Liu**, Huizi Mao, Jiarong Xing, Zirui Liu.
International Conference on Machine Learning (ICML), 2026.

RouterArena: An Open Platform for Comprehensive Comparison of LLM Routers

Yifan Lu*, **Rixin Liu***, Jiayi Yuan*, Xingqi Cui, Shenrun Zhang, Hongyi Liu, Jiarong Xing.
International Conference on Learning Representations (ICLR), 2026.

Who Routes the Router: Rethinking the Evaluation of LLM Routing Systems

Jiayi Yuan*, Yifan Lu*, **Rixin Liu**, Yu-Neng Chuang, Hongyi Liu, Shaochen Zhong, Yang Sui, Guanchu Wang, Jiarong Xing, Xia Hu.
NeurIPS Workshop on Evaluating the Evolving LLM Lifecycle, 2025.